

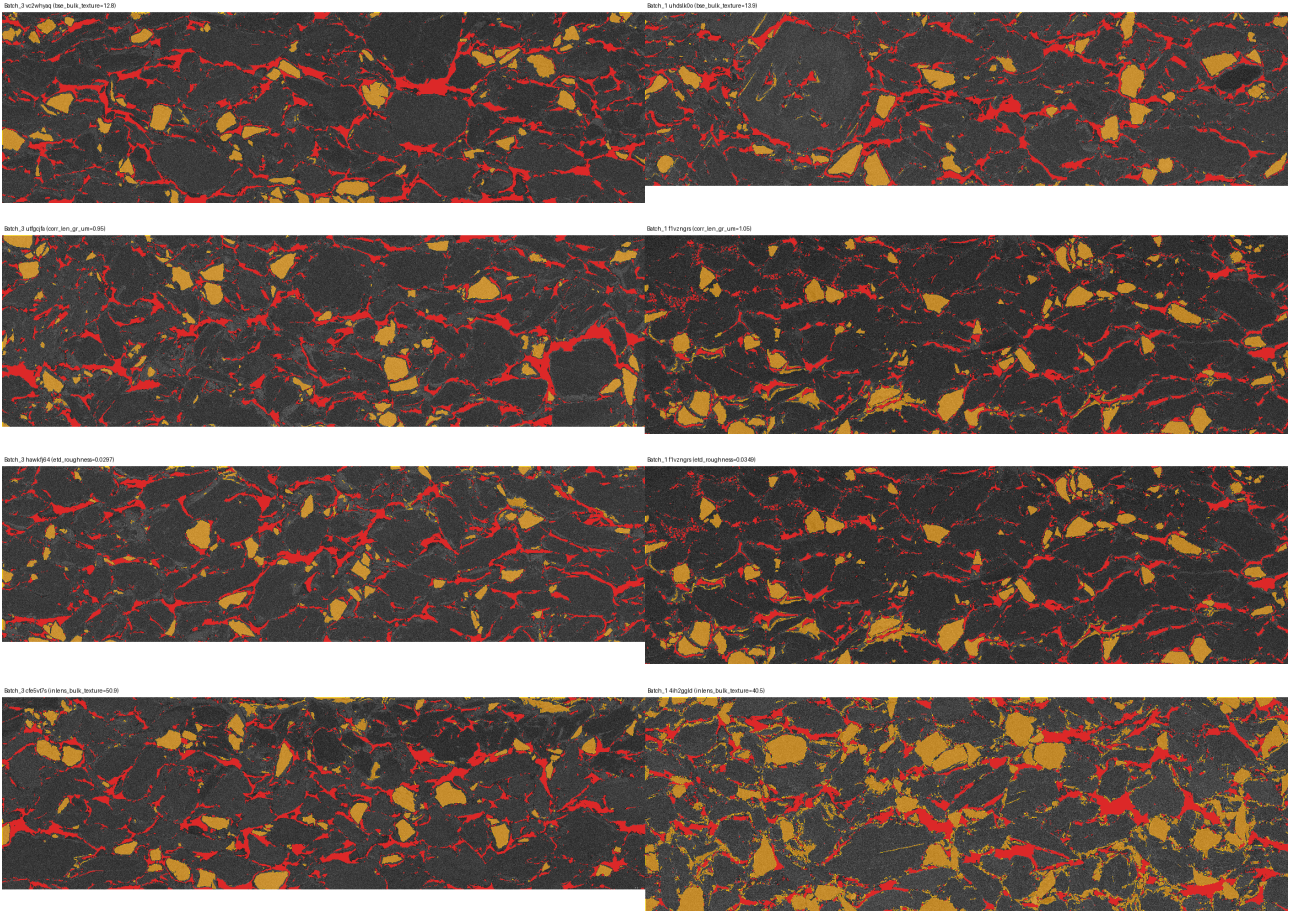
Phase 7 - Why the differences exist

Inference, not proof: Batch_1's lower visible porosity with fewer large gaps is best matched by denser packing / less mixing damage, NOT harder calendering - pore chords shrink isotropically whereas pressing predicts directional flattening. Both Batch_1 and Batch_2 carry a stronger imaging-noise signature (BSE bulk texture +15-18%, correlates 0.91 with the noise guard) - part of the between-batch signal is acquisition, not material. Alternative causes are listed and checked below; none is proven.

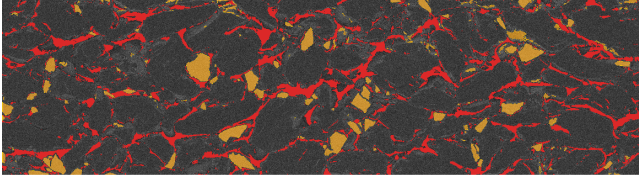
Assumption (applies throughout): the grey bulk is assumed graphite, bright particles are assumed silicon-based, black is pore/crack. This comes from image appearance and is unconfirmed.

	Predicts	Observed
	less porosity, fewer large gaps, FLATTER pores (H/V anisotropy)	pore_frac -13%, large_gap_frac -67%, but pore chords shrink isotropically (-5% both axes, anisotropy unchanged)
B1)	uniformly less pore space, fewer large voids, same Si size	pore_frac -13%, large_gap_frac -67%, crack_frac -13%, Si size equivalent
	different Si size/shape distribution	Si D10/D50 equivalent-to-undetermined
	changed Si clustering, graphite correlation length	Clark-Evans R +7% (undetermined), corr_len +18% (suggestive)
	global contrast and texture shifts	Si-bulk contrast lower in 2 flagged B1; BSE texture +16-18% tracks noise guard
	uniform noise/texture offset	noise_mad clusters: B1&B2 ~10.4 vs B3 ~7.4-10.4; bse_bulk_texture r=0.91 with noise

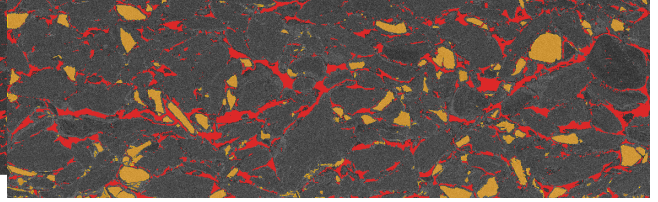
References opened: Chen et al. 2020 (JES 167 080534, open access) for electrode parameterisation; Profatilova et al. 2020 (ACS AEM, main text paywalled, free SI) for porosity/calendering effects; Torquato 2002 (Princeton-hosted paper) for two-point statistics; BoneJ 'thickness' docs for local thickness; Wikipedia 'Tortuosity' for tau definitions.



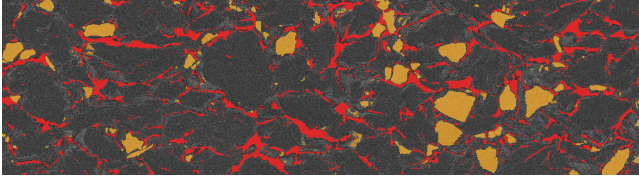
Batch_3_huax59f4 (large_gap, fac=0.0080)



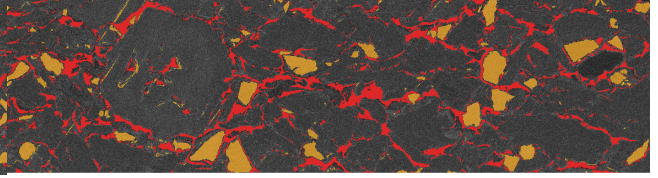
Batch_1_fz12d6f (large_gap, fac=0.00526)



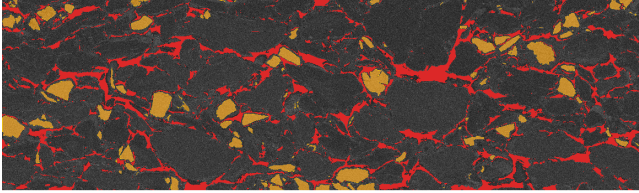
Batch_3_mgashp4 (one_fac=0.108)



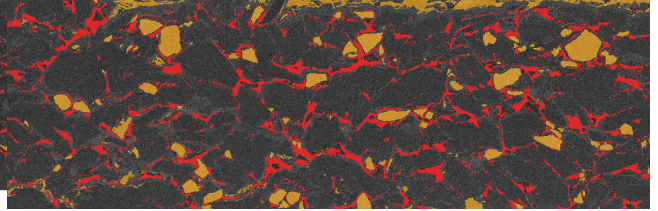
Batch_1_un98d6f (one_fac=0.104)



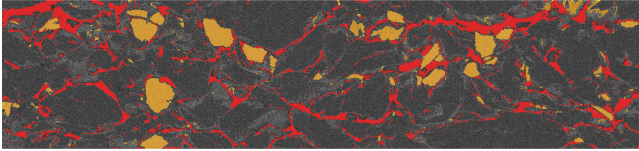
Batch_3_m2afp4 (one_fac=0.128)



Batch_3_fj1q9d (one_fac=0.138)



Batch_3_qj4f6 (gr_at_reference=0.119)



Batch_2_mq9d6f (gr_at_reference=0.0989)

